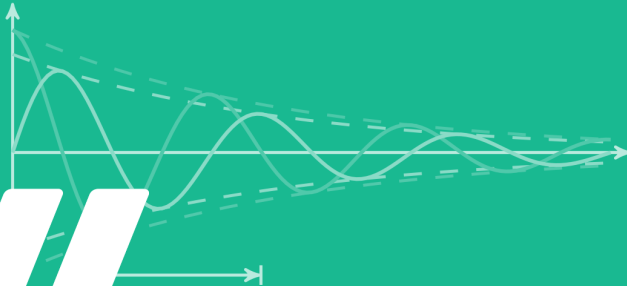


GET it digital

## Modul 8: Variable frequency circuits



Stand: 18. Februar 2026



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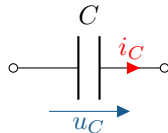
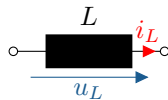
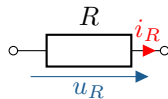
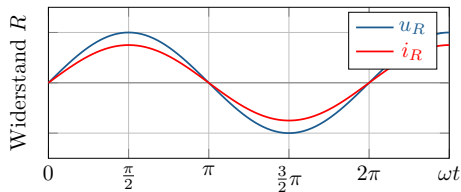
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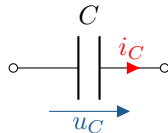
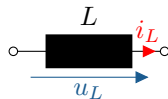
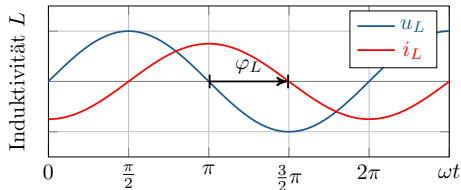
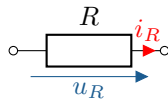
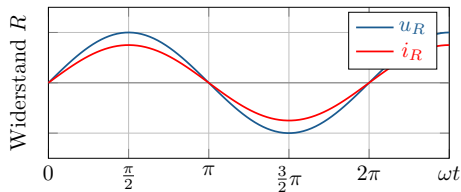
The work is available online at:

<https://getitdigital.uni-wuppertal.de/module/modul-8-schaltungen-variabler-frequenz>

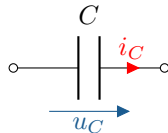
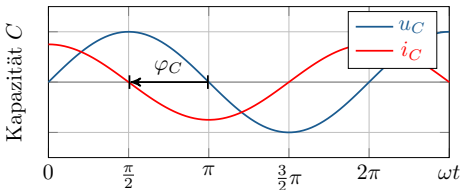
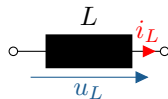
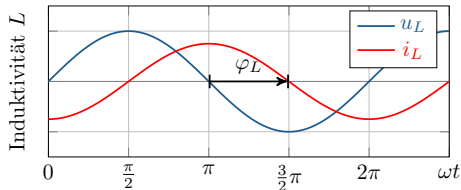
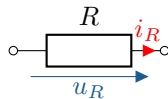
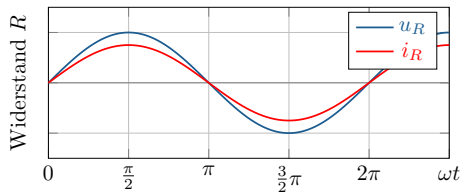
# Frequency dependence of electrical components



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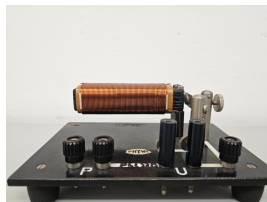
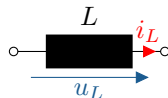


# Frequency dependence of electrical components



## Inductance

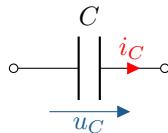
- ▶ Impedanz:  $\underline{Z}_L = j\omega L$
- ▶ DE:  $u_L = L \cdot \frac{d}{dt} i_L$
- ▶ Energy in the magnetic field
- ▶ Steady currents (no jumps)



Coil

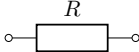
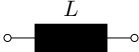
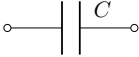
## Capacity

- ▶ Impedance:  $\underline{Z}_C = -j\frac{1}{\omega C}$
- ▶ DE:  $i_C = C \cdot \frac{d}{dt} u_C$
- ▶ Energy in the electric field
- ▶ Voltages steady (no jumps)



Capacitor

# Comparison of linear two-pole elements $R$ , $L$ and $C$

| Size            | Generally                                                                         | El. resistance                                                                    | Inductance                                                                          | Capacity                                                                            |
|-----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Symbol          |  |  |  |  |
| Unit            | [Form.s.] = Unit                                                                  | $[R] = \Omega$ (Ohm)                                                              | $[L] = \text{H}$ (Henry)                                                            | $[C] = \text{F}$ (Farad)                                                            |
| Time domain     | $\frac{d}{dt}$ or $\int dt$                                                       | $u_R = R \cdot i_R$                                                               | $u_L = L \cdot \frac{d}{dt} i_L$                                                    | $i_C = C \cdot \frac{d}{dt} u_C$                                                    |
| frequency range | $j\omega$ or $\frac{1}{j\omega}$                                                  | $\underline{U}_R = R \cdot \underline{I}_R$                                       | $\underline{U}_L = j\omega L \cdot \underline{I}_L$                                 | $\underline{I}_C = j\omega C \cdot \underline{U}_C$                                 |
| Impedance       | $\underline{Z} = \frac{\underline{U}}{\underline{I}}$                             | $\underline{Z}_R = R$                                                             | $\underline{Z}_L = j\omega L$                                                       | $\underline{Z}_C = -j\frac{1}{\omega C}$                                            |
| Active part     | $R = \text{Re}\{\underline{Z}\}$                                                  | $R = \frac{U}{I}$                                                                 | 0                                                                                   | 0                                                                                   |
| Blind part      | $X = \text{Im}\{\underline{Z}\}$                                                  | 0                                                                                 | $X_L = \omega L$                                                                    | $X_C = -\frac{1}{\omega C}$                                                         |

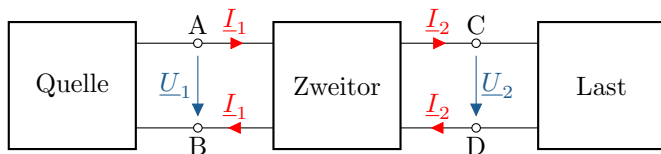
## Learning objectives: Frequency response, amplitude response, phase response

Students learn:

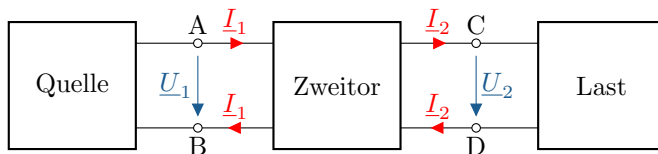
- ▶ Determine frequency responses algebraically based on circuit topology
- ▶ Understand how simple high-pass and low-pass filters work
- ▶ Analyse the limiting behaviour of four-poles based on their frequency responses



## Analysis of frequency-variable behaviour of electrical networks

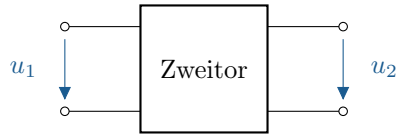


## Analysis of frequency-variable behaviour of electrical networks



**What is the transmission behavior like?**

## Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



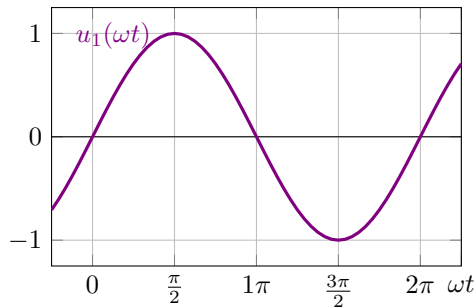
**Linear and time-invariant (LTI):**

## Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



### Linear and time-invariant (LTI):

- Sine waveform at the input  $u_1(\omega t)$

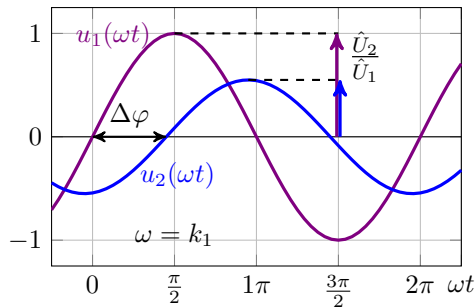


## Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



### Linear and time-invariant (LTI):

- ▶ Sine waveform at the input  $u_1(\omega t)$
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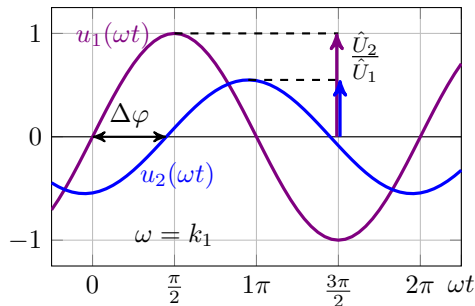


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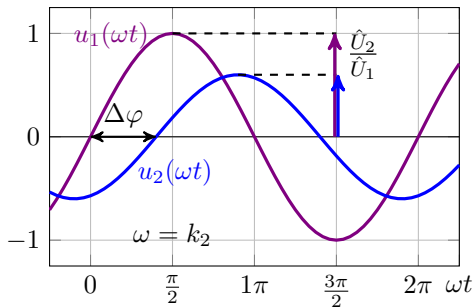


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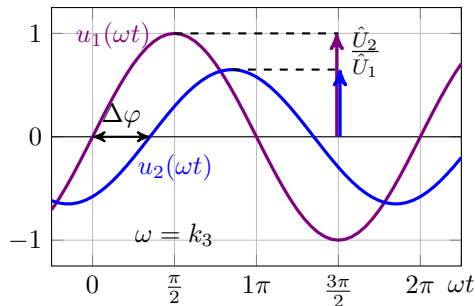


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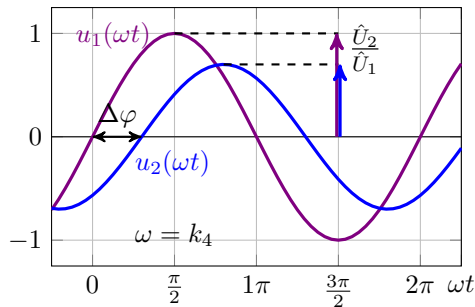


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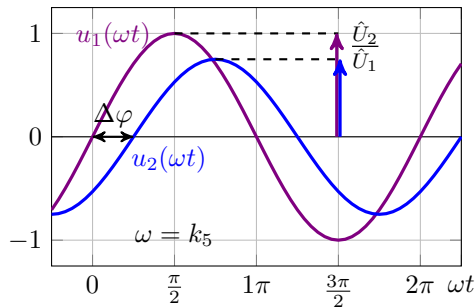


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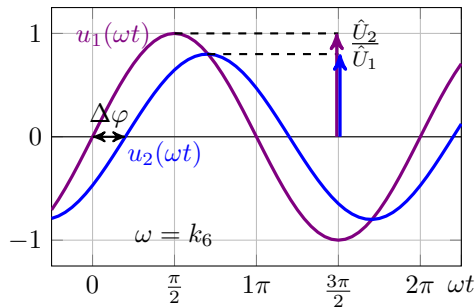


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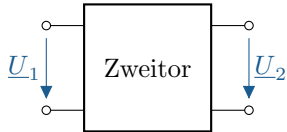
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# Definition of frequency response, amplitude response and phase response

## Secondary (four-pole):



# Definition of frequency response, amplitude response and phase response

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## Linear, time-invariant (LTI):

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

$$\xrightarrow{LZI} u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi)$$

Sine waveform is maintained

Frequency equal  $\omega_2 = \omega_1$

# Definition of frequency response, amplitude response and phase response

## Secondary (four-pole):



## Frequency response:

$$\underline{F}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = A(\omega) \cdot e^{j\varphi(\omega)}$$

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$$\begin{aligned} A(\omega) &= |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} \\ &= \sqrt{(\operatorname{Re}\{\underline{F}(j\omega)\})^2 + (\operatorname{Im}\{\underline{F}(j\omega)\})^2} \end{aligned}$$

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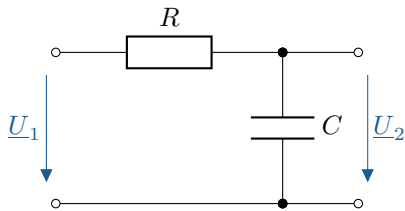
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## Phase shift:

$$\begin{aligned} \varphi(\omega) &= \angle \underline{F}(j\omega) = \angle \underline{U}_2 - \angle \underline{U}_1 \\ &= \arctan \left( \frac{\operatorname{Im}\{\underline{F}(j\omega)\}}{\operatorname{Re}\{\underline{F}(j\omega)\}} \right) \end{aligned}$$



# Frequency response using the example of a simple low-pass filter

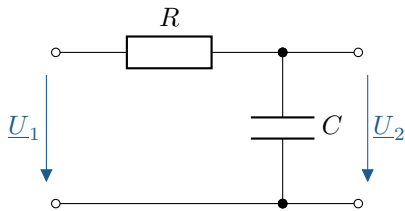


## 1st order RC low-pass filter

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

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# Frequency response using the example of a simple low-pass filter



## 1st order RC low-pass filter

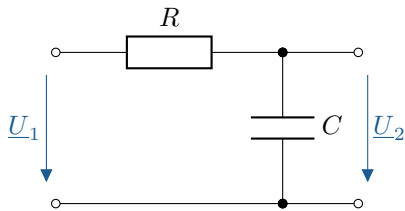
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## Frequency response:

$$\begin{aligned} \underline{F}(j\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{j\omega CR + 1} \\ &= \frac{1}{1 + j\omega CR} = \frac{1 - j\omega CR}{1 + (\omega CR)^2} \end{aligned}$$

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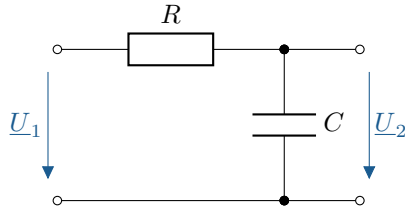
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## Amplitude and phase response:

$$A(\omega) = |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\varphi(\omega) = \arctan\left(\frac{\text{Im}\{\underline{F}\}}{\text{Re}\{\underline{F}\}}\right) = \arctan(-\omega CR)$$

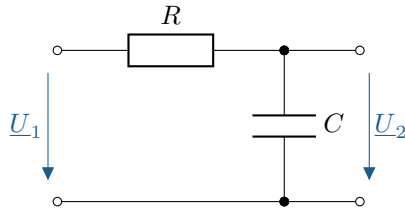
# Boundary behaviour using the example of a simple low-pass filter



**Limit behavior:** for  $f \rightarrow 0$  and  $f \rightarrow \infty$

$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 + j\omega CR}$$

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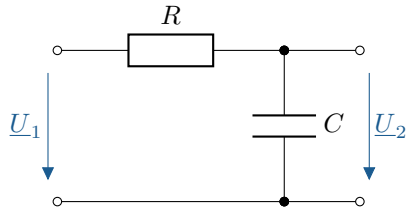


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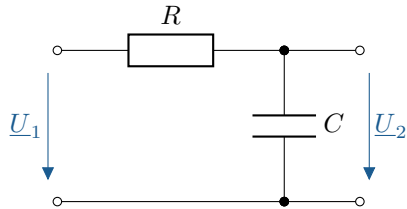
$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 + j\omega CR}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{1}{1} = 1$$

$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{\sqrt{\infty}} = 0$$

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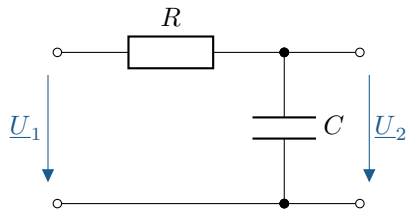
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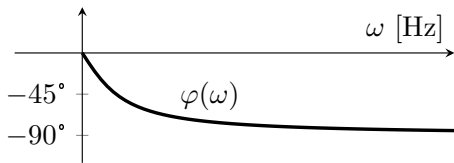
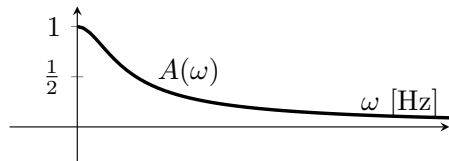
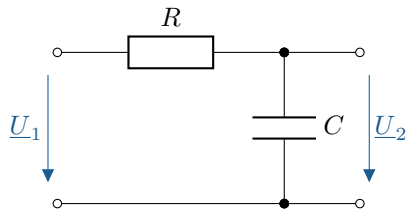
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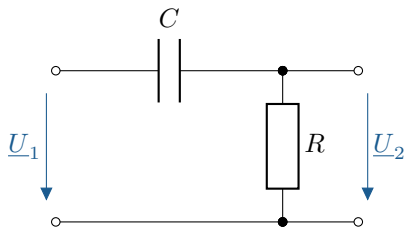
$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{\sqrt{\infty}} = 0$$

$$\varphi(\omega) = \arctan(-\omega CR)$$

$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(0) = 0^\circ$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$$

# Frequency response using the example of a simple high-pass filter



**1st order RC high pass filter**

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

$$\xrightarrow{LZI} u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi)$$

**Frequency response:**

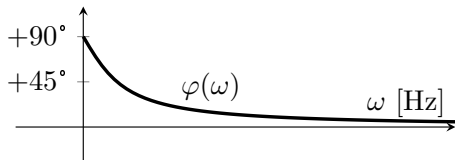
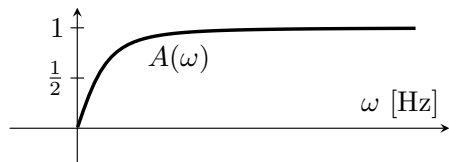
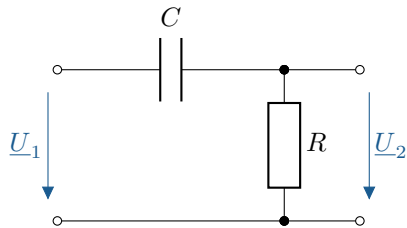
$$\begin{aligned} \underline{F}(j\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{1}{1 + \frac{1}{j\omega CR}} \\ &= \frac{1}{1 - j \frac{1}{\omega CR}} = \frac{1 + j \frac{1}{\omega CR}}{1 + \frac{1}{(\omega CR)^2}} \end{aligned}$$

**Amplitude and phase response:**

$$A(\omega) = |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} = \frac{1}{\sqrt{1 + \frac{1}{(\omega CR)^2}}}$$

$$\varphi(\omega) = \arctan\left(\frac{\text{Im}\{\underline{F}\}}{\text{Re}\{\underline{F}\}}\right) = \arctan\left(\frac{1}{\omega CR}\right)$$

# Boundary behaviour using the example of a simple high-pass filter



**Limit behaviour:** for  $f \rightarrow 0$  and  $f \rightarrow \infty$

$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 - j\frac{1}{\omega CR}}$$

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}}$$

$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{1}{\infty} = 0$$

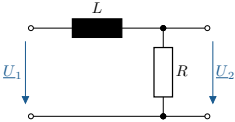
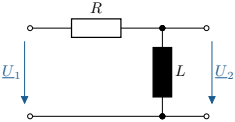
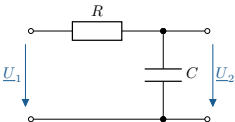
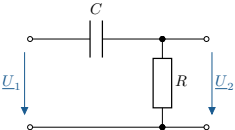
$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{1} = 1$$

$$\varphi(\omega) = \arctan\left(\frac{1}{\omega CR}\right)$$

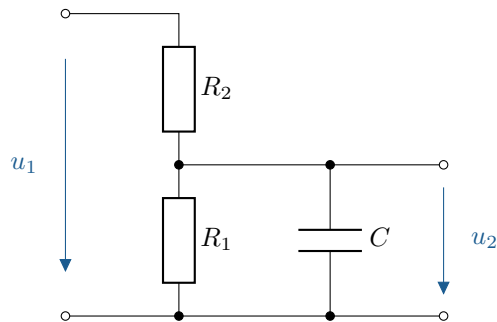
$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(+\infty) = +90^\circ$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(0) = 0^\circ$$

# Comparison of first-order passive filters with $L$ and $R$

|    | Low-pass filter                                                                   | $\underline{F}(j\omega)$            | High-pass filter                                                                   | $\underline{F}(j\omega)$             | $\omega_g$     |
|----|-----------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------|--------------------------------------|----------------|
| RL |  | $\frac{1}{1 + j\omega \frac{L}{R}}$ |  | $\frac{1}{1 - j\frac{R}{\omega L}}$  | $\frac{R}{L}$  |
| RC |  | $\frac{1}{1 + j\omega CR}$          |  | $\frac{1}{1 - j\frac{1}{\omega CR}}$ | $\frac{1}{CR}$ |

## Example filter circuit: Task



### Task:

- Derive  $\underline{F}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1}$  in general.
- Plot the magnitude and phase of  $\underline{F}(j\omega)$  for:

$$R_1 = 900 \, \Omega$$

$$R_2 = 100 \, \Omega$$

$$C = 1,25 \, \mu\text{F}$$

$$\begin{aligned} u_1(t) &= \hat{U}_1 \cdot \sin(\omega t) & \underline{U}_1 &= \hat{U}_1 \\ u_2(t) &= \hat{U}_2 \cdot \sin(\omega t + \varphi) & \underline{U}_2 &= \hat{U}_2 \cdot e^{j\varphi} \end{aligned}$$

- At what frequency  $f$  is  $\hat{U}_2 = \frac{1}{2}\hat{U}_1$ ?  
What phase shift does  $u_2$  now have relative to  $u_1$ ?

## Solution a) Frequency response in general

$$\begin{aligned}\underline{F}(j\omega) &= \frac{R_1 || \underline{Z}_C}{R_2 + (R_1 || \underline{Z}_C)} = \frac{\frac{\frac{R_1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}}{R_2 + \frac{\frac{R_1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}} \left| \cdot \frac{R_1 + \frac{1}{j\omega C}}{R_1 + \frac{1}{j\omega C}} \right. \\&= \frac{\frac{R_1}{j\omega C}}{R_1 \cdot R_2 + \frac{R_2}{j\omega C} + \frac{R_1}{j\omega C}} = \frac{R_1}{R_1 + R_2 + j\omega C R_1 R_2} \quad \hat{=} \frac{a + jb}{c + jd} \quad a, b, c, d \in \mathbb{R} \\&= \frac{R_1(R_1 + R_2) - j\omega C R_1^2 R_2}{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2} \quad (\text{Cartesian}) \quad \hat{=} \frac{a' + jb'}{c'} \quad a', b', c' \in \mathbb{R}\end{aligned}$$

## Example filter circuit: Task- Solution II

$$\underline{F}(j\omega) = \frac{R_1}{\underbrace{R_1 + R_2 + j\omega C R_1 R_2}_{\hat{=} \frac{a+jb}{c+jd}}} = \frac{R_1(R_1 + R_2) - j\omega C R_1^2 R_2}{\underbrace{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2}_{\hat{=} \frac{a'+jb'}{c'}}$$

Amount:  $|A(\omega)| = \frac{R_1}{\sqrt{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2}}$  with  $|\underline{F}(j\omega)| = \frac{\sqrt{a^2 + b^2}}{\sqrt{c^2 + d^2}}$

Phase:  $\varphi(\omega) = \arctan\left(\frac{-\omega C R_1^2 R_2}{R_1(R_1 + R_2)}\right)$  with  $\varphi(\omega) = \arctan\left(\frac{b'}{a'}\right)$

$$= \arctan\left(\frac{-\omega C R_1 R_2}{R_1 + R_2}\right)$$

Altern.:  $\varphi(\omega) = \arctan(0) - \arctan\left(\frac{\omega C R_1 R_2}{R_1 + R_2}\right)$  with  $\varphi(\omega) = \underbrace{\arctan(\frac{b}{a})}_{\varphi_Z} - \underbrace{\arctan(\frac{d}{c})}_{\varphi_N}$

## Solution b) Limit behaviour and sketch

$$A(\omega) = \frac{R_1}{\sqrt{(R_1+R_2)^2 + (\omega C R_1 R_2)^2}}$$

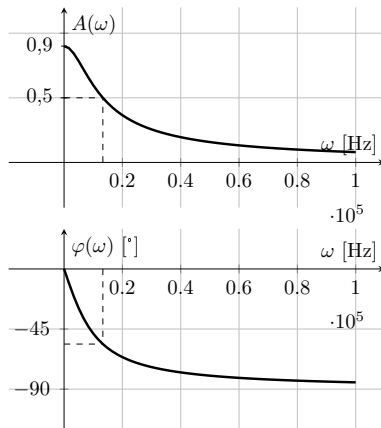
$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{R_1}{R_1+R_2} = 0,9$$

$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{R_1}{\infty} = 0$$

$$\varphi(\omega) = \arctan\left(-\frac{\omega C R_1 R_2}{R_1+R_2}\right)$$

$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(0) = 0$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$$





## Solution c) Determine specific operating point

Total: Frequency  $f_1$  where  $\hat{U}_2 = \frac{1}{2}\hat{U}_1$  and phase shift  $u_2$  relative to  $u_1$  at  $f_1$ :

$$A(\omega_1) = \frac{R_1}{\sqrt{(R_1 + R_2)^2 + (\omega CR_1R_2)^2}} \stackrel{!}{=} \frac{1}{2} \quad \text{with} \quad A(\omega) = \frac{U_2}{U_1} = \frac{\hat{U}_2}{\hat{U}_1} \stackrel{!}{=} \frac{1}{2} \Big|_{f=f_1}$$

$$\Rightarrow \frac{(R_1 + R_2)^2 + (\omega_1 CR_1R_2)^2}{R_1^2} = 4 \quad \Rightarrow \quad (R_1 + R_2)^2 + (\omega_1 CR_1R_2)^2 = 4R_1^2$$

$$\Rightarrow (\omega_1 CR_1R_2)^2 = 4R_1^2 - (R_1 + R_2)^2 \quad \Rightarrow \quad \omega_1 = \pm \frac{\sqrt{4R_1^2 - (R_1^2 + 2R_1R_2 + R_2^2)}}{CR_1R_2}$$

$$\omega_1 = 1,3304 \cdot 10^4 \frac{1}{s}, f_1 = 2,117 \text{ kHz} \quad \text{with} \quad \omega_1 < 0 \text{ unphysical, } f = \frac{\omega}{2\pi}$$

$$\varphi(\omega_1) = \arctan(-1,4967) = -56,25^\circ \quad \text{with} \quad \varphi(\omega) = \arctan\left(\frac{\text{Im}\{\underline{F}\}}{\text{Re}\{\underline{F}\}}\right)$$

## Learning objectives: Logarithmische Darstellung, Frequenzgang

Students learn:

- ▶ Know the differences between linear and logarithmic representation.
- ▶ Know and use decibels as a unit of measurement.
- ▶ Know and determine the cut-off frequencies of filter circuits.
- ▶ Construct and interpret Bode diagrams.
- ▶ Know and understand how bandpass and band-stop filters work.
- ▶ Know and understand the structure and behaviour of higher-order filters.

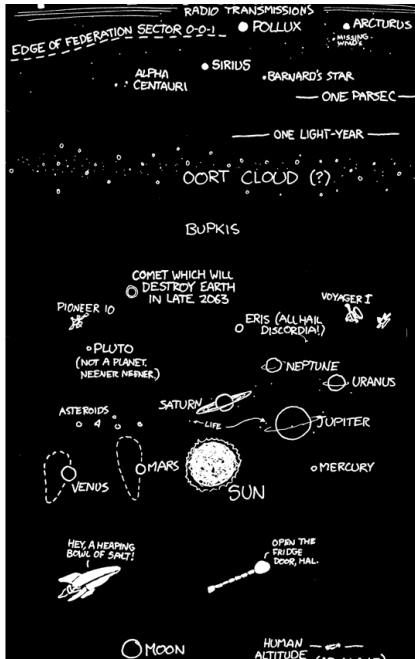
# Logarithmic representation, frequency response, cut-off frequency



Multipage Comic Strip (vertikal Scrollen)

Quelle: xkcd #482, *Depth*, <https://xkcd.com/485/> (Abruf: 04.04.2024)

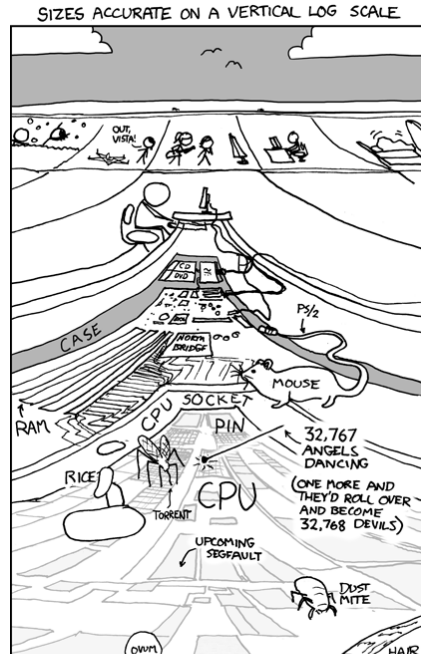
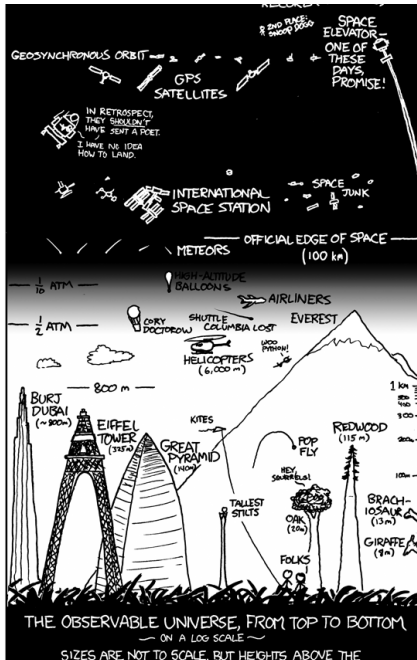
# Logarithmic representation, frequency response, cut-off frequency



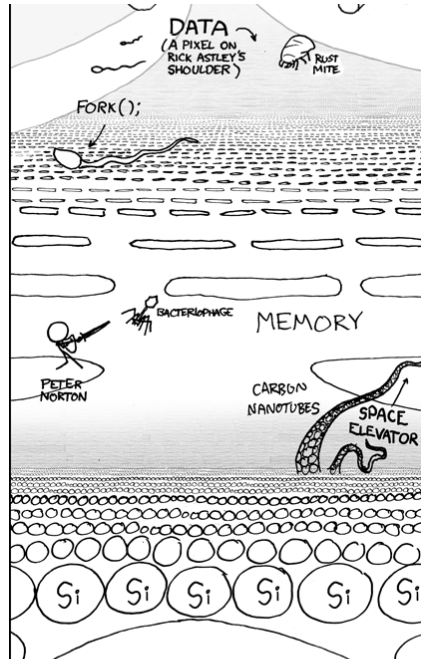
Multipage Comic Strip (vertikal Scrollen)

Quelle: xkcd #482, *Depth*, <https://xkcd.com/485/> (Abruf: 04.04.2024)

# Logarithmic representation, frequency response, cut-off frequency

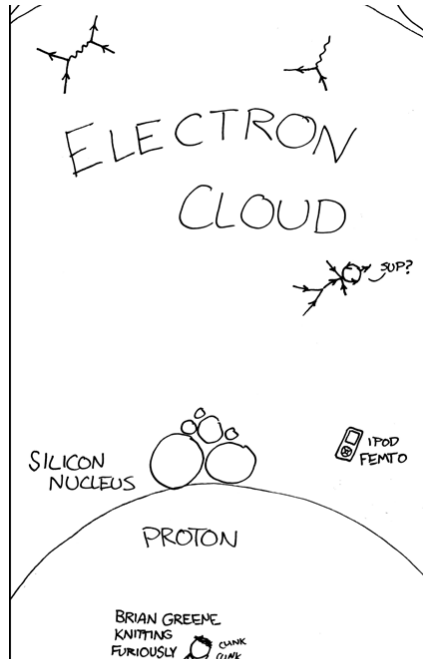


# Logarithmic representation, frequency response, cut-off frequency



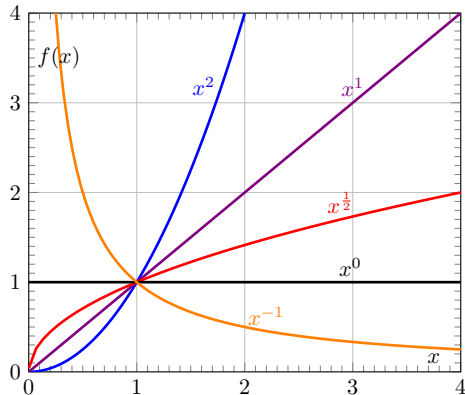
Quelle: xkcd #482, Height, <https://xkcd.com/482/> (Abruf: 04.04.2024)

# Logarithmic representation, frequency response, cut-off frequency



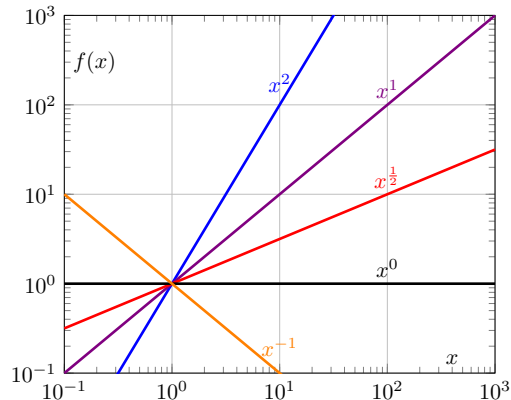
Quelle: xkcd #482, Height, <https://xkcd.com/482/> (Abruf: 04.04.2024)

# Logarithmic representation of power functions



## Linear representation

- ▶ Straight  $f(x) = a(x - x_0) + y_0$
- ▶ Add./Subtr.  $\rightarrow$  Shift
- ▶ Mult./Div.  $\rightarrow$  Stretching



## (Double logarithmic representation

- ▶ Straight:  $f(x) = (x/x_0)^a \cdot y_0$
- ▶ Mult./Div.  $\rightarrow$  shift
- ▶ Exponent  $\rightarrow$  Stretching



**Decibel** Auxiliary unit used to indicate logarithmic ratios between two quantities.

Power indicators (z.B.  $P$ ,  $W$ )       $Q_{(P)} = \log \frac{P_2}{P_1}$  B =  $10 \cdot \log \frac{P_1}{P_2}$  dB

Power root sizes (z.B.  $U$ ,  $I$ )       $Q_{(R)} = \log \frac{U_2^2}{U_1^2}$  B =  $20 \cdot \log \frac{U_1}{U_2}$  dB

Typical decibel values (amplification in dB) at voltage ratios

| $\frac{U_2}{U_1}$ | $\frac{1}{\sqrt{2}}$ | 1    | $\sqrt{2}$ | 2     | $\sqrt{10}$ | 10    | 20    | 100   |
|-------------------|----------------------|------|------------|-------|-------------|-------|-------|-------|
| $Q_{dB}$          | -3 dB                | 0 dB | +3 dB      | +6 dB | 10 dB       | 20 dB | 26 dB | 40 dB |

**Cut-off frequency:**  $f_g$  (,  $\omega_g$ )

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

**Cut-off frequency:**  $f_g$  (,  $\omega_g$ )

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

**Sections:**

Passage area:  $A(\omega) > 1/\sqrt{2}$

Cutoff frequency:  $A(\omega_g) = 1/\sqrt{2}$

Blocked area:  $A(\omega) < 1/\sqrt{2}$

# Cut-off frequency of simple filter circuits

**Cut-off frequency:**  $f_g$  (,  $\omega_g$ )

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

**Sections:**

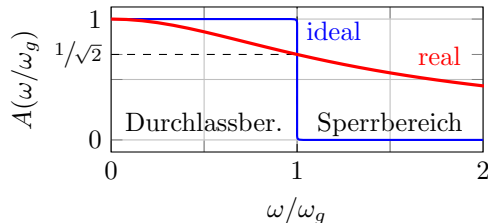
Passage area:  $A(\omega) > 1/\sqrt{2}$

Cutoff frequency:  $A(\omega_g) = 1/\sqrt{2}$

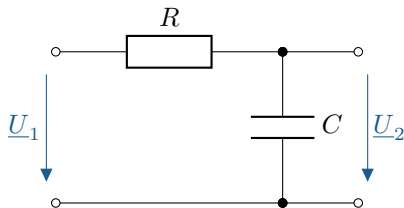
Blocked area:  $A(\omega) < 1/\sqrt{2}$

**Example:** First-order low-pass filter

$$A(\omega/\omega_g) = \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} \quad \text{mit } \omega_g = \frac{1}{CR}$$



## Logarithmic representation



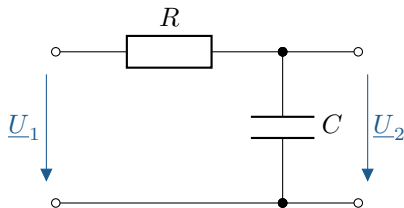
**1st order RC low-pass filter**

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

## Logarithmic representation



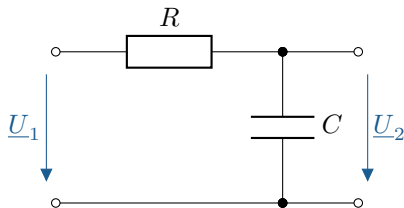
**1st order RC low-pass filter**

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

## Logarithmic representation



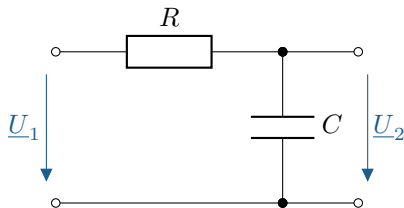
**1st order RC low-pass filter**

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC} \quad \omega_g = \frac{1}{RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

## Logarithmic representation



**1st order RC low-pass filter**

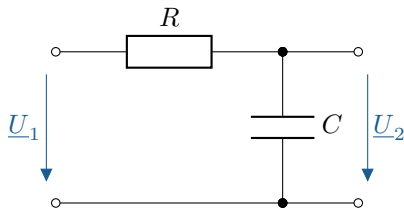
$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC} \quad \omega_g = \frac{1}{RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$

$$\varphi(\omega) = -\arctan(\omega RC) \quad \varphi(\omega_g) = -45^\circ$$



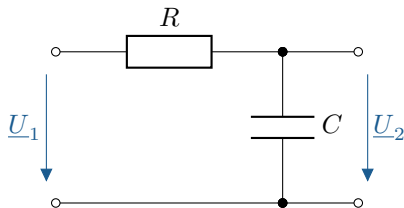
## Logarithmic representation



**1st order RC low-pass filter**

$$\begin{aligned} \underline{F}(j\omega) &= \frac{1}{1 + j\omega/\omega_g} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega/\omega_g) & \varphi(\omega_g) &= -45^\circ \end{aligned}$$

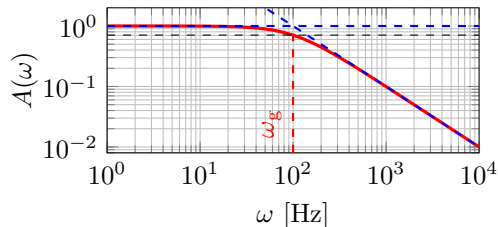
# First-order low-pass filter



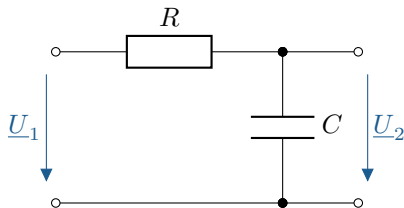
**1st order RC low-pass filter**

$$\begin{aligned} F(j\omega) &= \frac{1}{1 + j\omega/\omega_g} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega/\omega_g) & \varphi(\omega_g) &= -45^\circ \end{aligned}$$

**Logarithmic representation**



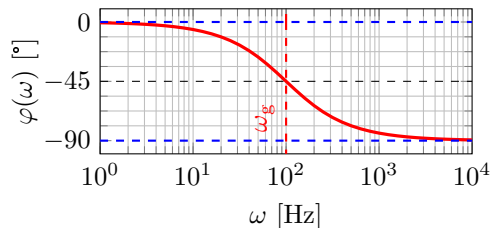
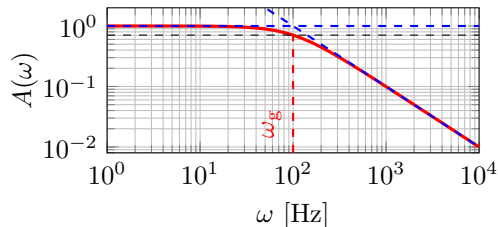
# First-order low-pass filter



**1st order RC low-pass filter**

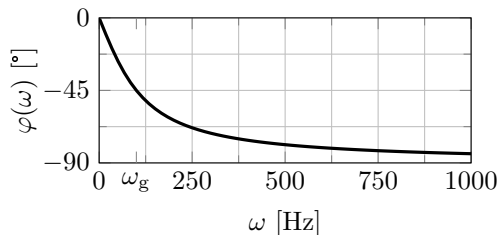
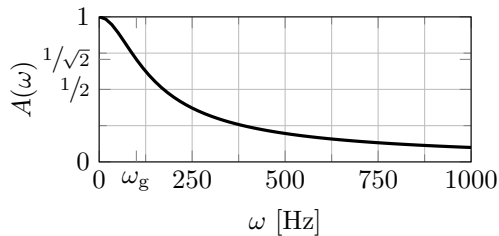
$$\underline{F}(j\omega) = \frac{1}{1 + j\omega/\omega_g} \quad \omega_g = \frac{1}{RC}$$
$$A(\omega) = \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$
$$\varphi(\omega) = -\arctan(\omega/\omega_g) \quad \varphi(\omega_g) = -45^\circ$$

**Logarithmic representation**

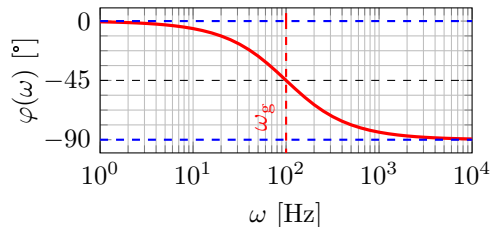
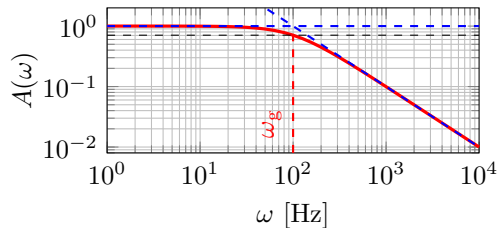


# Comparison of linear and logarithmic representations

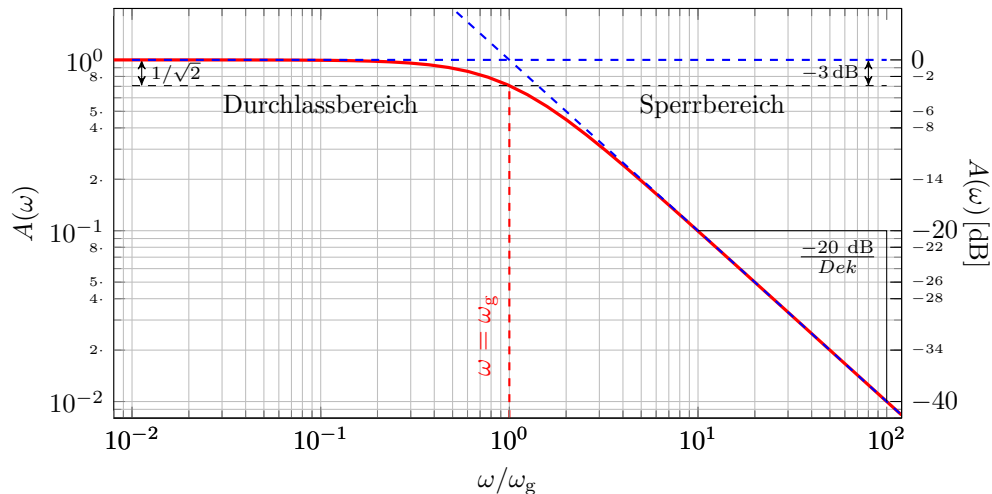
## Lineare Darstellung



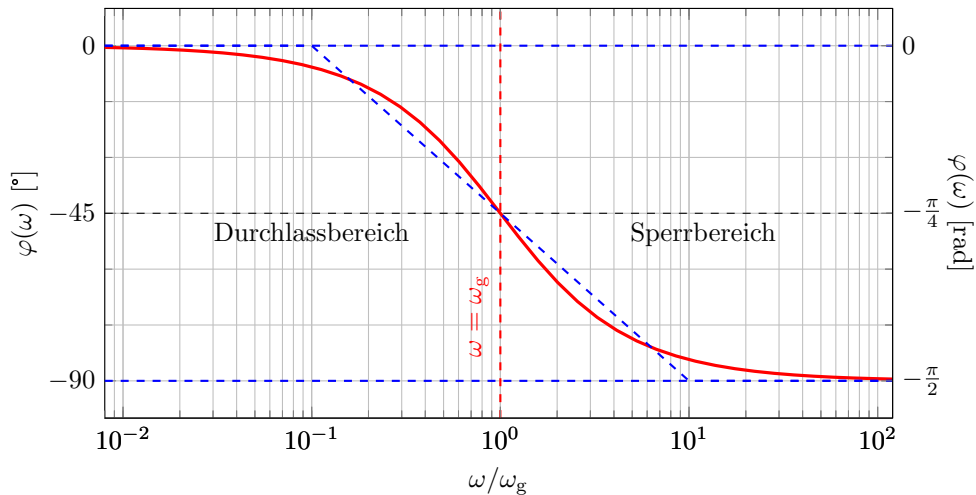
## Logarithmische Darstellung



# First-order low-pass filter, amplitude response

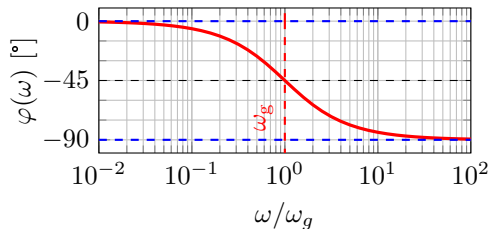
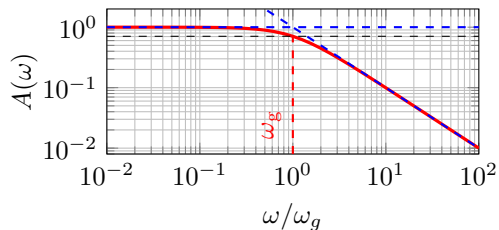


# Tiefpass 1. Ordnung, Phasengang

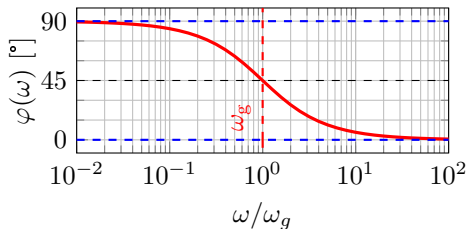
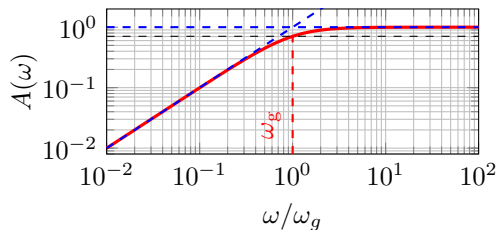


# Comparison of first-order low-pass and high-pass filters

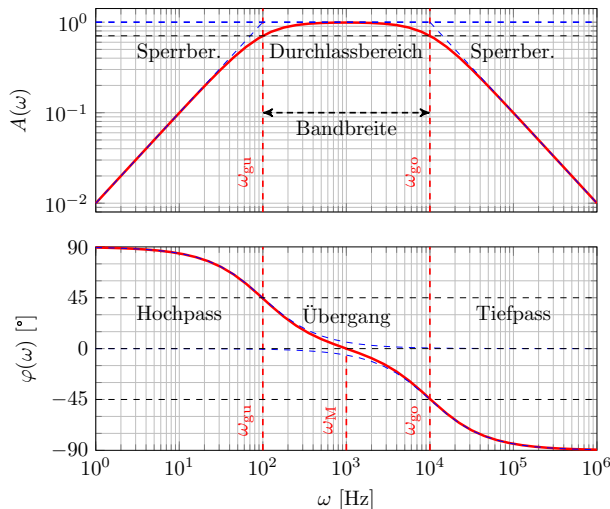
## Low-pass filter



## High-pass filter

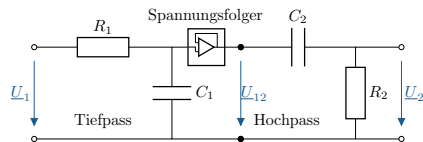


# Second-order bandpass filter



Frequency response<sup>1</sup>:  $\underline{F}_{BP}(j\omega) = \underline{F}_{TP}(j\omega) \cdot \underline{F}_{TP}(j\omega)$

<sup>1</sup>For non-reactive series connection from multiplication of individual frequency responses



Cutoff frequency: (upper, lower)

$$f_{go}, f_{gu}$$

Bandwidth:

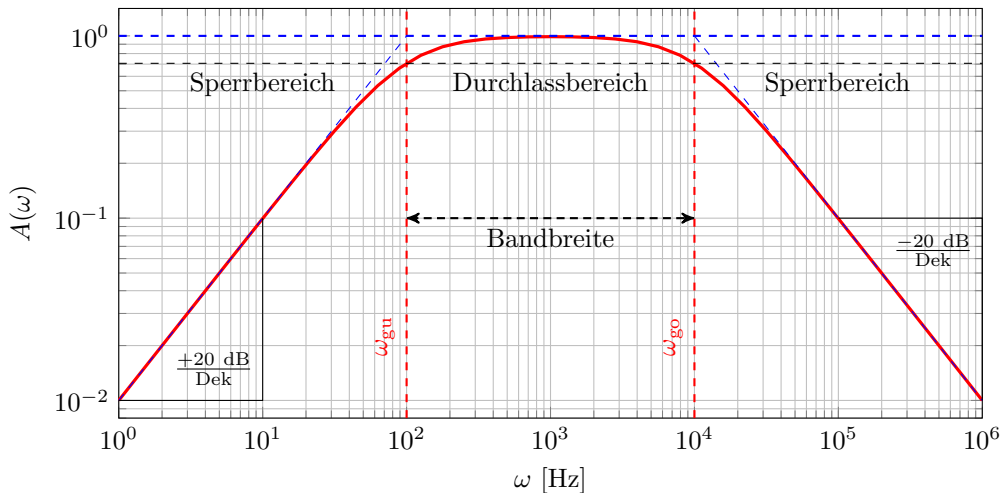
$$B = \Delta f = f_{go} - f_{gu}$$

Center frequency:

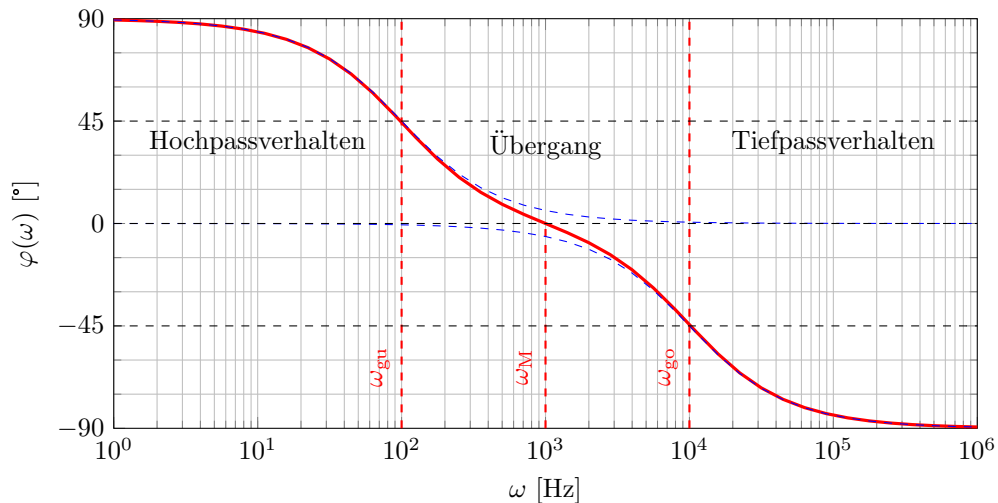
$$f_m = \sqrt{f_{go} \cdot f_{gu}}$$



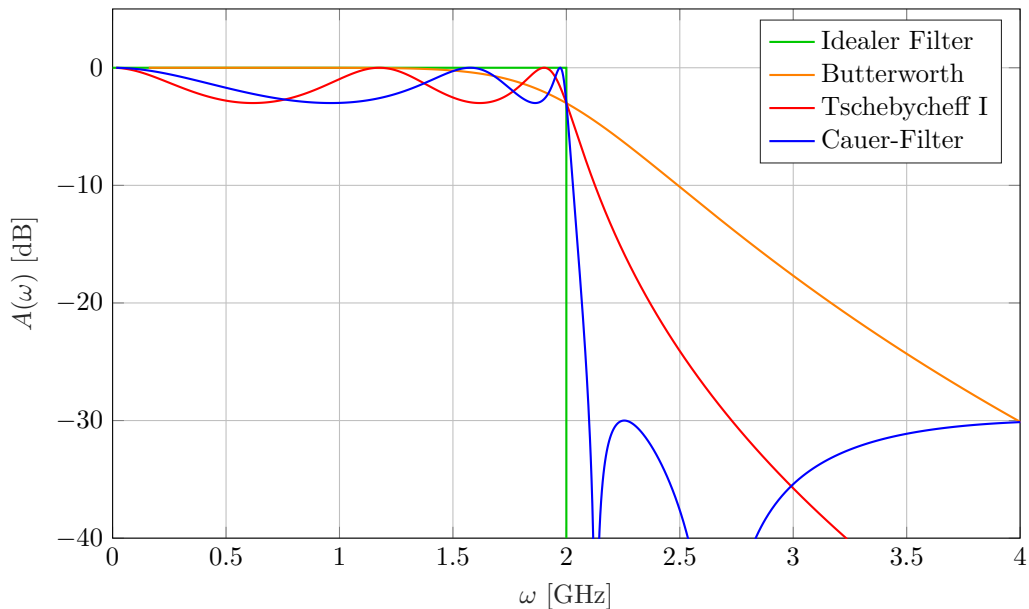
## Second-order bandpass filter – amplitude response, detail



## Bandpass 2. Ordnung - Phasengang, Detail



## Order of filters, example filter of higher order



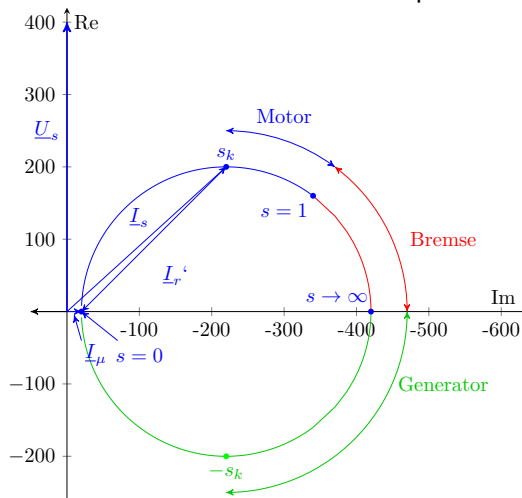
## Learning objectives: Locus curve

Students learn:

- ▶ Know locus curves and possible areas of application.
- ▶ Construct and interpret locus curves in principle.
- ▶ Know impedance and admittance locus curves of basic circuits.

**Def. Local curve:** Representation of a complex pointer  $\underline{z}(p) = \text{Re}\{\underline{z}(p)\} + j \text{Im}\{\underline{z}(p)\}$  at a complex level depending on a real parameter  $p$ .

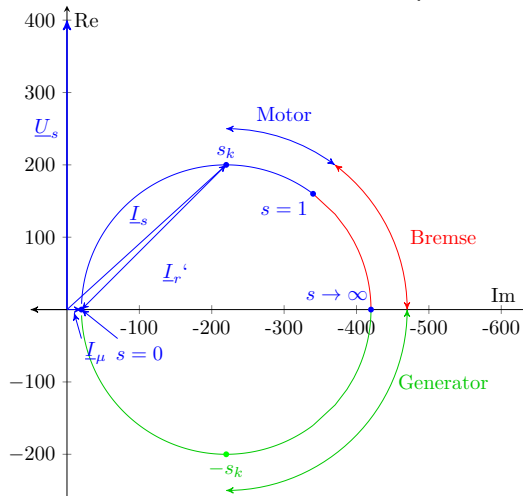
**Def. Local curve:** Representation of a complex pointer  $\underline{z}(p) = \text{Re}\{\underline{z}(p)\} + j \text{Im}\{\underline{z}(p)\}$  at a complex level depending on a real parameter  $p$ .



**Example:** Current-frequency curve of an asynchronous machine for  $\underline{I}_s(s)$

## Definition of location curve

**Def. Local curve:** Representation of a complex pointer  $\underline{z}(p) = \text{Re}\{\underline{z}(p)\} + j \text{Im}\{\underline{z}(p)\}$  at a complex level depending on a real parameter  $p$ .

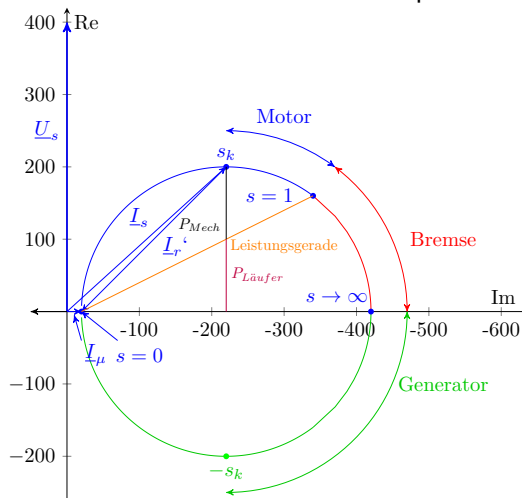


## Areas of application?

**Example:** Current-frequency curve of an asynchronous machine for  $\underline{I}_s(s)$

# Definition of location curve

**Def. Local curve:** Representation of a complex pointer  $\underline{z}(p) = \text{Re}\{\underline{z}(p)\} + j \text{Im}\{\underline{z}(p)\}$  at a complex level depending on a real parameter  $p$ .



**Areas of application?**

**Visualization of:**

- System states
- System sizes
- Influence of system sizes

**Example:** Current-frequency curve of an asynchronous machine for  $\underline{I}_s(s)$



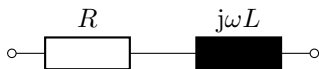
Pointer diagram:

Shows magnitude in the complex plane as a pointer (of a point).

## Key point: Location curve

Displays size in a complex plane as a curve (set of points) depending on parameters.

# Impedance locus curve - RL element, $R$ variable

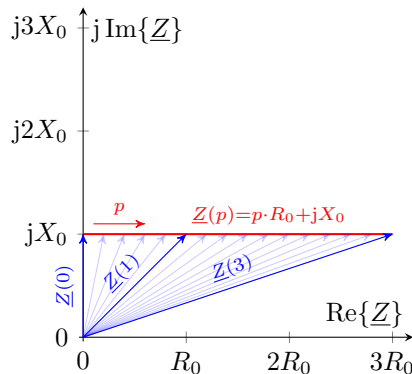


$$\underline{Z} = R + j\omega L$$

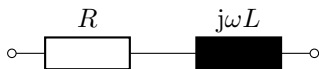
$$\underline{Z} = p \cdot R_0 + jX_0$$

with  $p \in [0,3]$

$R_0, X_0 = \text{const.}$



# Impedance locus curve - RLs element, $\omega$ variable

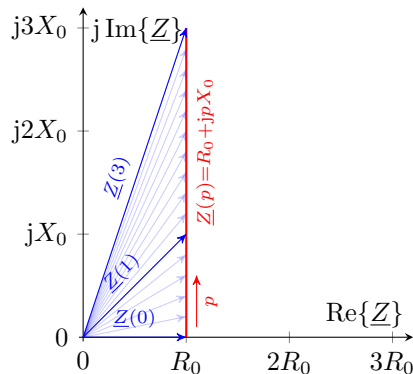


$$\underline{Z} = R + j\omega L$$

$$\underline{Z} = R_0 + jp \cdot \omega_0 L_0$$

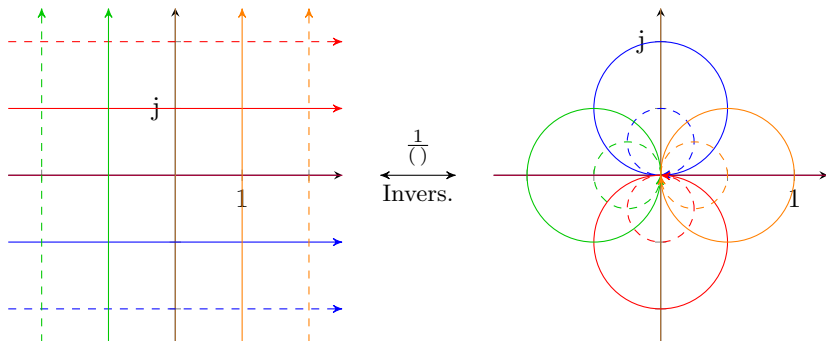
mit  $p \in [0,3]$

$R_0, L_0, \omega_0 = konst.$



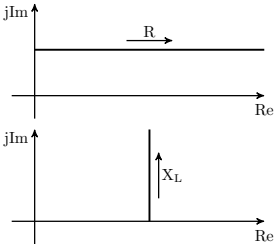
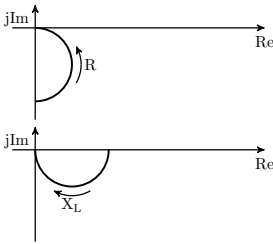
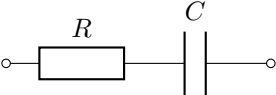
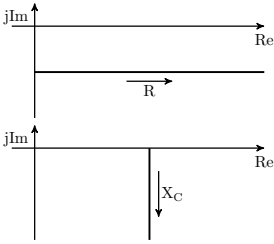
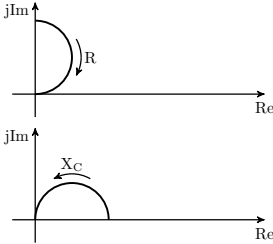
# Admittance locus curve - Inversion of locus curves

| original local curve |                        | inverted locus curve |                        |
|----------------------|------------------------|----------------------|------------------------|
| Line                 | not through the origin | Line                 | not through the origin |
|                      | Not through the origin | Circle               | through the origin     |
| Circle               | through the origin     | Line                 | not through the origin |
| Circle               | not through the origin | Circle               | not through the origin |

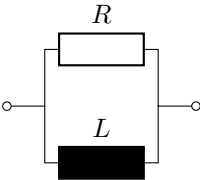
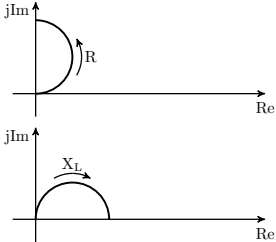
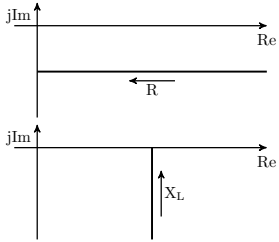
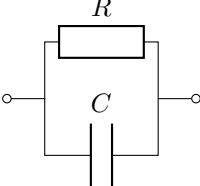
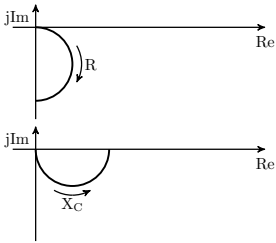
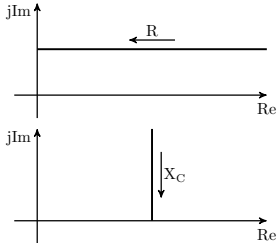


<sup>1</sup>Based on MoebiusInversion.svg by Chrislb, CC-BY-SA-2.0-DE, 2005,  
<https://commons.wikimedia.org/wiki/File:MoebiusInversion.svg>

# Local curves of basic circuits - in Series

| Basic-circuit                                                                     | <u>Z</u> -Locus-curve                                                              | <u>Y</u> -Locus-curve                                                               |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
|  |  |  |

# Local curves of basic circuits - in Parallel

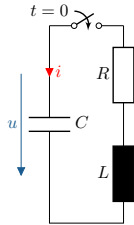
| Grundschialtung                                                                   | <u>Z</u> -Ortskurve                                                                | <u>Y</u> -Ortskurve                                                                 |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
|  |  |  |

## Learning objectives: Resonant circuits

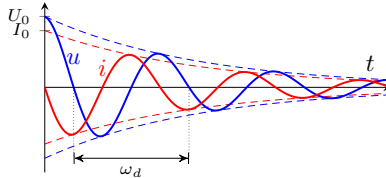
Students learn:

- ▶ Identifying and calculating resonance circuits
- ▶ Determine resonance frequency, characteristic impedance and quality factor
- ▶ Calculate, describe and plot resonance curves

# Resonance phenomenon



Free oscillation:

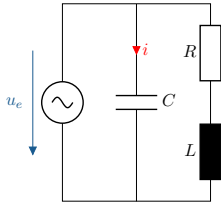


Boundary condition:

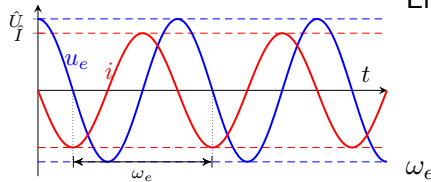
$$u(t = 0) = U_0$$

$$i(t = 0) = 0$$

with natural frequency  $\omega_d$ ,  
subsiding (subdued)



Forced oscillation:



Energizing voltage

$$u_e(t) = \hat{U} \sin(\omega_e t)$$

with Energizing frequency

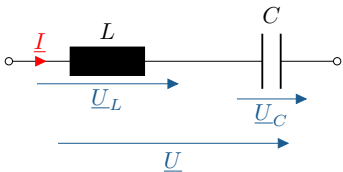
$\omega_e$



# Definition of resonance condition, frequency, quality factor, characteristic impedance

Grundtypen

Series resonant circuit



**Resonance condition**  $\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0$

**Resonance frequency**  $f_0$

**Characteristic impedance**  $X_k = |X_{L,0}| = |X_{C,0}|$

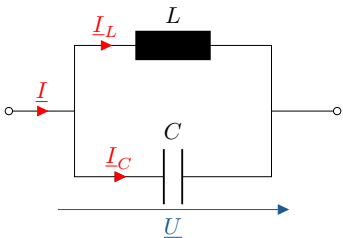
**Characteristic value**  $B_k = \frac{1}{X_k}$

**Quality factor**  $Q = \frac{|Q_{L,0}|}{P_0} = \frac{|Q_{C,0}|}{P_0}$

**Natural frequency**  $f_d$  (damped)

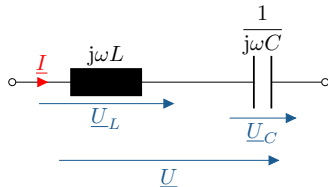
**Natural frequency**  $f_0$  (undamped)

Parallel resonant circuit



# Resonant frequency using the example of ideal LC oscillating circuits

in Series

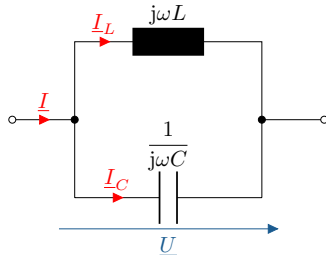


voltage-related

$$\begin{aligned}\underline{U} &= \underline{I} \cdot \left( j\omega L + \frac{1}{j\omega C} \right) \\ &= \underline{I} \cdot j \left( \underbrace{\omega L - \frac{1}{\omega C}}_{=0 \text{ Resonance}} \right)\end{aligned}$$

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\Rightarrow \underline{I} \rightarrow \infty \quad \text{für} \quad \underline{U} = \text{const.}$$



parallel

electricity-related

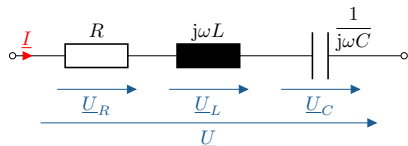
$$\begin{aligned}\underline{I} &= \underline{U} \cdot \left( j\omega C + \frac{1}{j\omega L} \right) \\ &= \underline{U} \cdot j \left( \underbrace{\omega C - \frac{1}{\omega L}}_{=0 \text{ Resonance}} \right)\end{aligned}$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$

$$\Rightarrow \underline{U} \rightarrow \infty \quad \text{für} \quad \underline{I} = \text{const.}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

# RLC series resonant circuit - resonance behaviour



$$\underline{Z} = R + j \underbrace{\left( \omega L - \frac{1}{\omega C} \right)}_{= 0 \text{ for } \omega = \omega_0}$$

## Resonance condition (allg.)

Phase shift between  $\underline{U}$  and  $\underline{I}$  corresponds to  $\varphi = 0$ .

Or:

$$\operatorname{Im} \left\{ \frac{\underline{U}}{\underline{I}} \right\} = \operatorname{Im} \{ \underline{Z} \} \stackrel{!}{=} 0$$

Case 1:  $\omega < \omega_0$

$$\rightarrow \left( \omega L - \frac{1}{\omega C} \right) < 0 \quad \text{ohmic-capacitive}$$

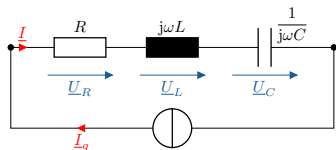
Case 2:  $\omega = \omega_0$  (Resonance case)

$$\rightarrow \left( \omega L - \frac{1}{\omega C} \right) = 0 \quad \text{purely ohmic}$$

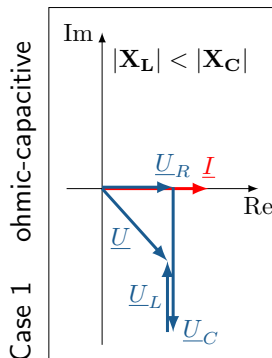
Case 3:  $\omega > \omega_0$

$$\rightarrow \left( \omega L - \frac{1}{\omega C} \right) > 0 \quad \text{ohmic-inductive}$$

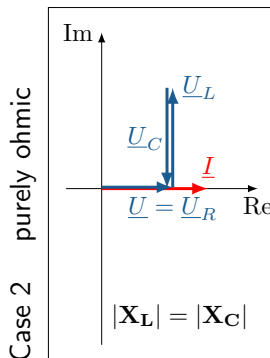
# Case differentiation in the pointer diagram



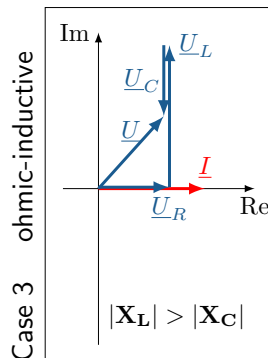
$$\begin{aligned}\underline{U} &= \underline{U}_R + \underline{U}_L + \underline{U}_C & \underline{I} &= \text{const.} \\ &= \underline{I} \cdot \left[ R + j \underbrace{\left( \omega L - \frac{1}{\omega C} \right)}_{= \text{Im}\{\underline{Z}\}} \right] & \omega_0 &= \frac{1}{\sqrt{LC}}\end{aligned}$$



$$\omega < \omega_0$$

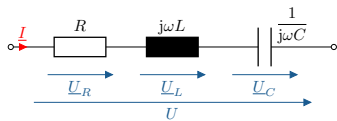


$$\omega = \omega_0$$



$$\omega > \omega_0$$

# Case differentiation in impedance curve



$$\underline{Z} = R + j \left( \omega L - \frac{1}{\omega C} \right)$$

Resonance condition

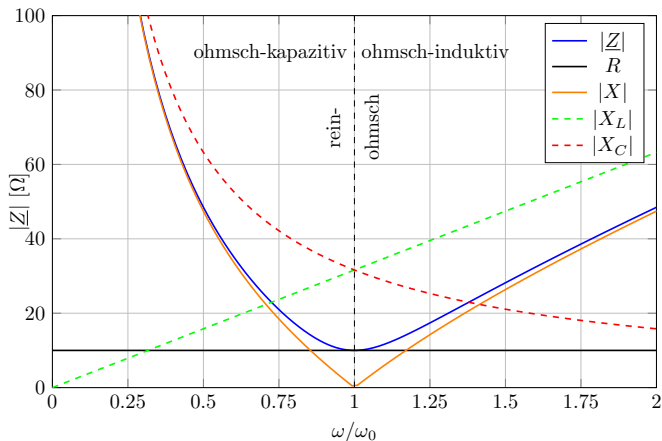
$$\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0$$

Resonance frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

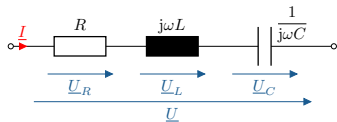
Resonant impedance

$$\underline{Z}_0 = R$$



$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

# Resonance curve and quality factor of an RLC series resonant circuit



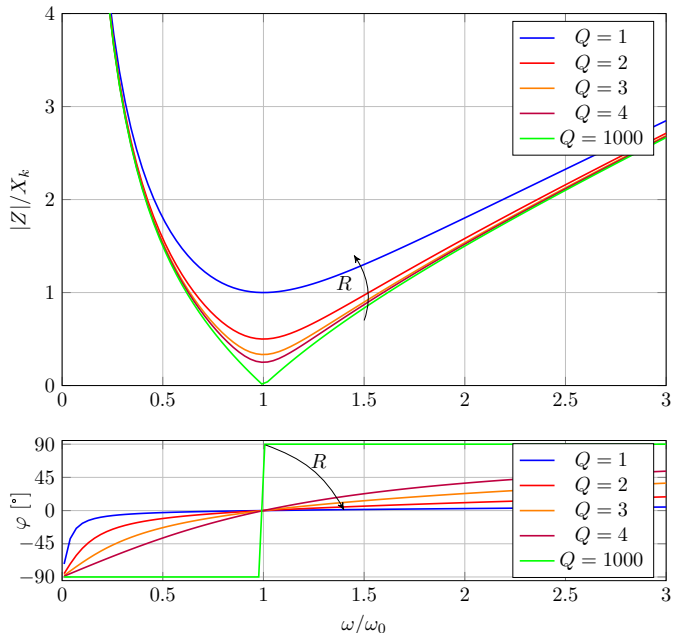
$$\underline{Z} = R + j \left( \omega L - \frac{1}{\omega C} \right)$$

standardized

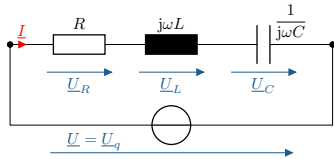
$$\frac{\underline{Z}}{X_k} = \frac{1}{Q} + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

Quality and char. resistance

$$Q = \frac{X_k}{R} \quad X_k = \sqrt{\frac{L}{C}}$$



# Voltage resonance using the example of an RLC series resonant circuit I

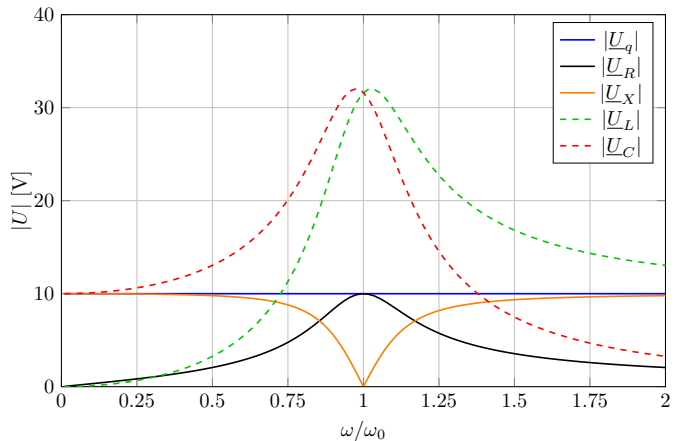


Voltage divider (e.g.)

$$\frac{U_L}{U} = \frac{j \left( \frac{\omega}{\omega_0} \right)}{\frac{1}{Q} + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

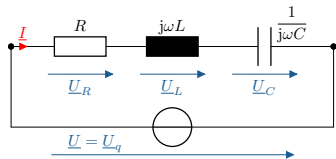
Quality and overvoltage

$$Q = \frac{X_k}{R} = \frac{U_{L,0}}{U} = \frac{U_{C,0}}{U}$$



$U_q(\omega_0) \Rightarrow \text{Overvoltage!}$

# Additional calculation



$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = \underline{U}_q$$

$$\underline{I} = \underline{I}_R = \underline{I}_L = \underline{I}_C$$

$$\underline{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

$$Q = \frac{U_{L,0}I}{U_{R,0}I} = \frac{U_{L,0}}{U_q} = \frac{X_k}{R}$$

$$\frac{\underline{U}_R}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{R}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$\frac{\underline{U}_L}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{j\omega L}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$\frac{\underline{U}_C}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{\frac{1}{j\omega C}}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

standardized

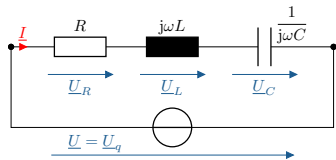
$$\frac{\underline{U}_R}{\underline{U}} = \frac{1/Q}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\frac{\underline{U}_L}{\underline{U}} = \frac{+j\frac{\omega}{\omega_0}}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\frac{\underline{U}_C}{\underline{U}} = \frac{-j\frac{\omega_0}{\omega}}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$



# Current curve of the RLC series resonant circuit II

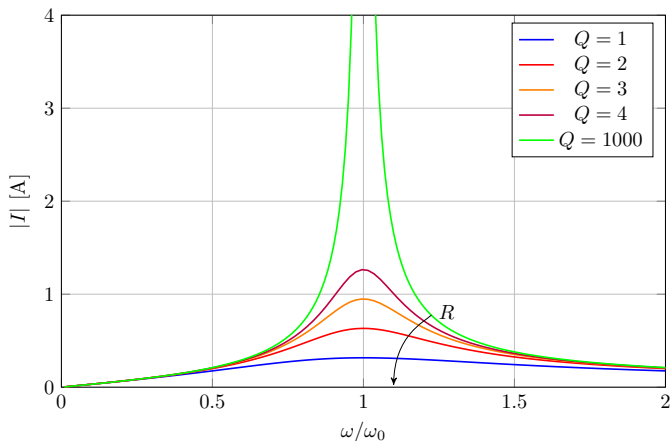


Current

$$\underline{I} = \frac{I_0}{1 + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) Q}$$

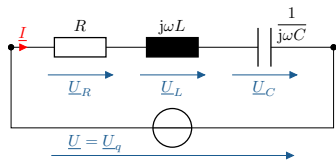
Maximum current, Quality factor

$$I_0 = \frac{U_q}{R} \quad Q = \frac{X_k}{R}$$



Maximum current at  $\omega_0$   
Limitation by  $R$

## Additional calculation



$$\underline{Z} = R + j(X_L + X_C)$$

$$X_L = +X_k \frac{\omega}{\omega_0} = \omega L$$

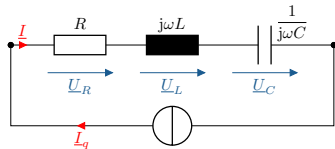
$$X_C = -X_k \frac{\omega_0}{\omega} = -\frac{1}{\omega C}$$

$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

$$Q = \frac{X_k}{R} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\begin{aligned} \underline{I} &= \frac{\underline{U}_q}{\underline{Z}} \\ &= \frac{\underline{U}_q}{R + j\left(\omega L - \frac{1}{\omega C}\right)} \\ &= \frac{\underline{U}_q}{R + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k} \\ &= \frac{\underline{U}_q}{R} \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \frac{X_k}{R}} \\ &= \underline{I}_0 \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q} \end{aligned}$$

# Voltage curve of the RLC series resonant circuit



Current source

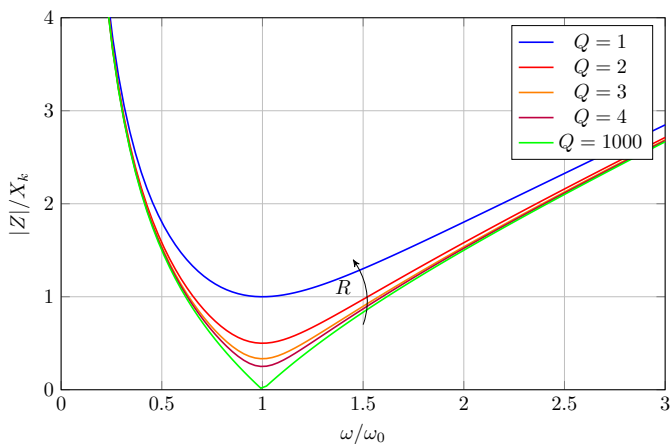
$$\underline{I} = I_q$$

Voltage

$$\underline{U} = I_q \cdot \underline{Z}$$

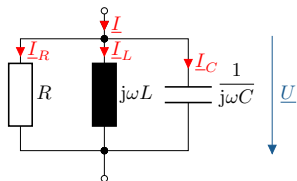
Impedance

$$\underline{Z} = R + jX$$



Voltage curve  $\sim$  Impedance curve

# RLC parallel resonant circuit - resonance behavior



$$\underline{Y} = G + j \underbrace{\left( \omega C - \frac{1}{\omega L} \right)}_{= 0 \text{ for } \omega = \omega_0}$$

## Resonance condition (gener.)

Phase shift between  $\underline{U}$  and  $\underline{I}$  corresponds to  $\varphi = 0$ .

Or:

$$\operatorname{Im} \left\{ \frac{\underline{I}}{\underline{U}} \right\} = \operatorname{Im} \{ \underline{Y} \} \stackrel{!}{=} 0$$

Case 1:  $\omega < \omega_0$

$$\rightarrow \left( \omega C - \frac{1}{\omega L} \right) < 0 \quad \text{ohmic-inductive}$$

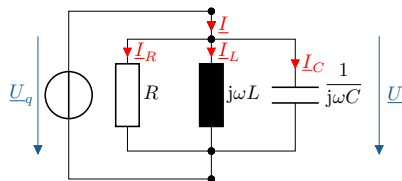
Case 2:  $\omega = \omega_0$  (Resonance case)

$$\rightarrow \left( \omega C - \frac{1}{\omega L} \right) = 0 \quad \text{purely ohmic}$$

Case 3:  $\omega > \omega_0$

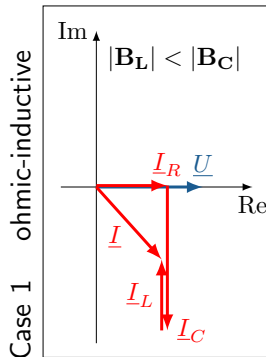
$$\rightarrow \left( \omega C - \frac{1}{\omega L} \right) > 0 \quad \text{ohmic-capacitive}$$

# Case differentiation in the pointer diagram

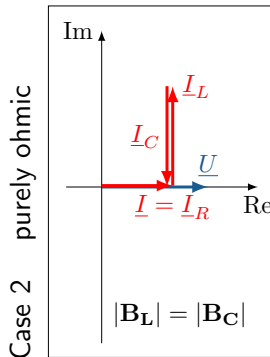


$$\begin{aligned}\underline{I} &= \underline{I}_R + \underline{I}_L + \underline{I}_C \\ &= \underline{U} \cdot \left[ G + j \left( \omega C - \frac{1}{\omega L} \right) \right] \\ &= \underline{U} \cdot \underbrace{\left[ G + j \left( \omega C - \frac{1}{\omega L} \right) \right]}_{= \text{Im}\{\underline{Y}\}}\end{aligned}$$

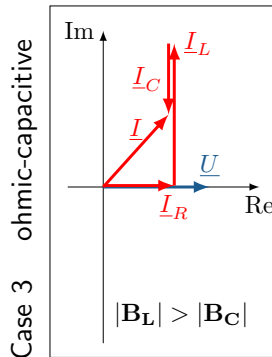
$$\begin{aligned}\underline{U} &= \text{const.} \\ \omega_0 &= \frac{1}{\sqrt{LC}}\end{aligned}$$



$$\omega < \omega_0$$

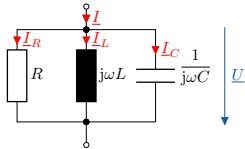


$$\omega = \omega_0$$



$$\omega > \omega_0$$

# Case differentiation in admittance curve



$$\underline{Y} = G + j \left( \omega C - \frac{1}{\omega L} \right)$$

Resonance condition

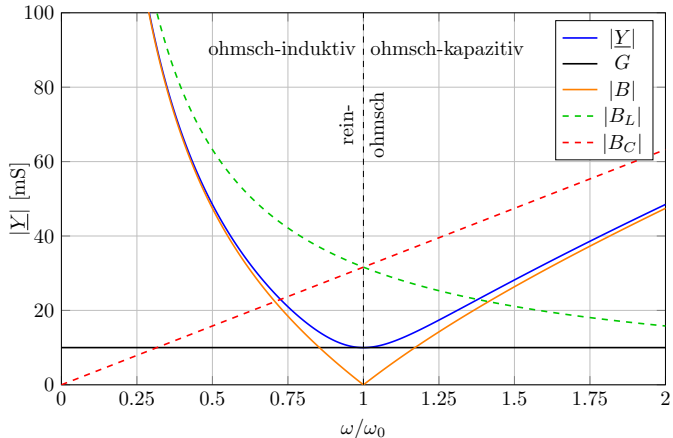
$$\text{Im}\{\underline{Y}\} \stackrel{!}{=} 0$$

Resonance frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

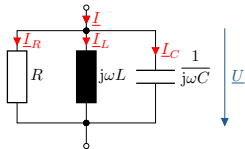
Resonance admittance

$$\underline{Y}_0 = G$$



$$B_k = |B_{C,0}| = |B_{L,0}| = \sqrt{\frac{C}{L}}$$

# Resonance curve and quality factor of an RLC parallel resonant circuit



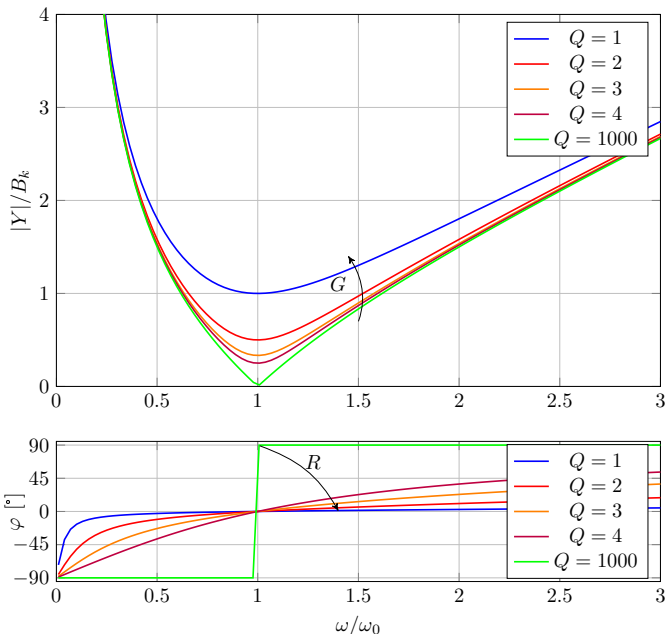
$$\underline{Y} = G + j \left( \omega C - \frac{1}{\omega L} \right)$$

standardized

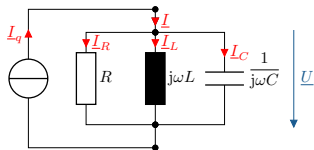
$$\frac{\underline{Y}}{B_k} = \frac{1}{Q} + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

Quality factor and char. value

$$Q = \frac{B_k}{G} \quad B_k = \sqrt{\frac{C}{L}}$$



# Current resonance using the example of an RLC parallel resonant circuit



$$\underline{U} = \underline{U}_R = \underline{U}_L = \underline{U}_C$$

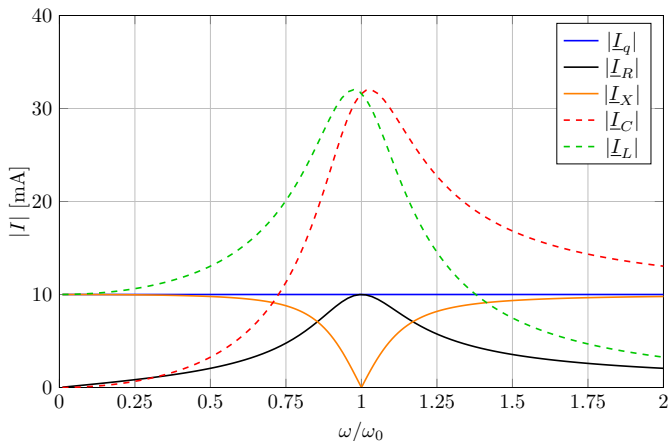
$$\underline{I} = \underline{I}_R + \underline{I}_L + \underline{I}_C = \underline{I}_q$$

Current divider (Bsp.)

$$\frac{\underline{I}_C}{\underline{I}} = \frac{j \left( \frac{\omega}{\omega_0} \right)}{\frac{1}{Q} + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

Quality factor and over-current

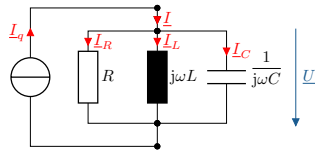
$$Q = \frac{B_k}{\omega} = \frac{I_{C,0}}{I} = \frac{I_{L,0}}{I}$$



Current source  $I_q(\omega_0) \Rightarrow$  Overcurrent!



# Voltage curve of the RLC parallel resonant circuit II

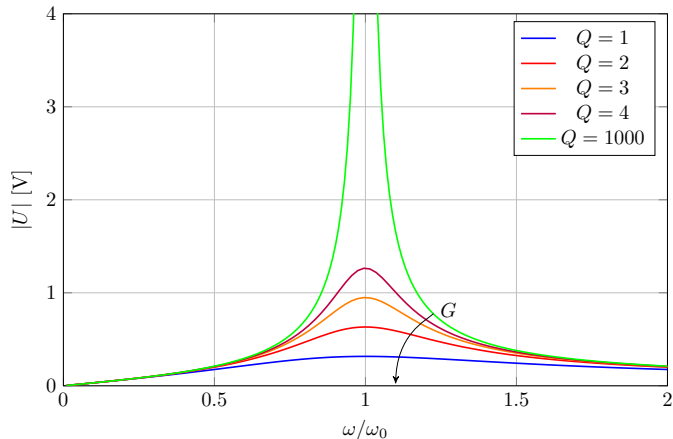


Voltage

$$\underline{U} = \frac{U_0}{1 + j \left( \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) Q}$$

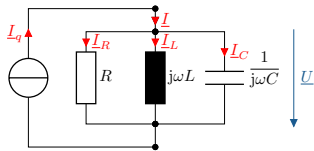
Maximum voltage, Quality factor

$$U_0 = \frac{I_q}{G} \quad Q = \frac{B_k}{G}$$



Maximum voltage at  $\omega_0$   
Limitation by  $G$

# Additional calculation



$$\underline{Y} = G + j(B_L + B_C)$$

$$B_L = -B_k \frac{\omega_0}{\omega} = -\frac{1}{\omega L}$$

$$B_C = +B_k \frac{\omega}{\omega_0} = \omega C$$

$$B_k = |B_{L,0}| = |X_{C,0}| = \sqrt{\frac{C}{L}}$$

$$Q = \frac{B_k}{G} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{U} = \frac{\underline{I}_q}{\underline{Y}}$$

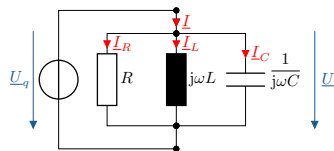
$$= \frac{\underline{I}_q}{G + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$= \frac{\underline{I}_q}{G + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k}$$

$$= \frac{\underline{I}_q}{G} \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \frac{B_k}{G}}$$

$$= \underline{U}_0 \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q}$$

# Current curve of the RLC parallel resonant circuit



Voltage source

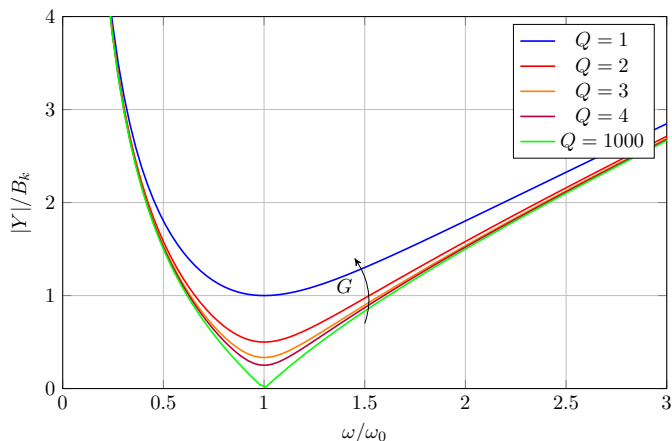
$$\underline{U} = U_q$$

Current

$$\underline{I} = U_q \cdot \underline{Y}$$

Admittance

$$\underline{Y} = G + jB$$



Current curve  $\sim$  Admittance curve

